

MEAN, MEDIAN, MODE

MEASURES OF DISPERSION

Formula cheatsheet with individual, discrete, and continuous examples

Exam Focus.

Notation. x = observation or class mark; f = frequency; $N = \sum f$ = total frequency; L = lower class boundary; h = class width; $c.f.$ = cumulative frequency before the relevant class.

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1. Data types and notation

Data type	Meaning	What to prepare
Individual / ungrouped	Each value is listed separately.	Ordered values for median and quartiles.
Discrete frequency	Values x have frequencies f .	Columns x , f , fx , cumulative f .
Continuous frequency	Values are grouped into class intervals.	Class boundaries, class marks, f , and cumulative f .

Remember.

For continuous data, use the class mark $x = (\text{lower boundary} + \text{upper boundary})/2$ for mean, mean deviation, and standard deviation. The result is an approximation because all observations in a class are represented by its midpoint.

2. Measures of central tendency

2.1 Arithmetic mean

Data	Formula
Individual	$\bar{x} = \frac{\sum x}{N}$
Discrete frequency	$\bar{x} = \frac{\sum fx}{\sum f} = \frac{\sum fx}{N}$
Continuous frequency	$\bar{x} = \frac{\sum fx}{N}$, where x is the class mark.

Shortcut / assumed-mean method

$$\bar{x} = A + \frac{\sum fd}{N}, \quad d = x - A.$$

For equal class width, step-deviation method:

$$\bar{x} = A + h \frac{\sum fu}{N}, \quad u = \frac{x - A}{h}.$$

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Mean examples.

Individual: For 2, 4, 6, 8, 10, 12, $\bar{x} = 42/6 = 7$.

Discrete: If $x = 10, 20, 30, 40$ with $f = 2, 3, 4, 1$, then $N = 10$, $\sum fx = 240$, so $\bar{x} = 240/10 = 24$.

Continuous: For classes 0–10, 10–20, 20–30, 30–40 with $f = 2, 3, 4, 1$, class marks are 5, 15, 25, 35. Thus $\bar{x} = (10 + 45 + 100 + 35)/10 = 19$.

2.2 Median

Median is the middle value after arranging data in ascending order.

Data	Formula / method
Individual, n odd	Position = $(n + 1)/2$.
Individual, n even	Median = $\frac{(n/2)\text{th value} + ((n/2) + 1)\text{th value}}{2}$.
Discrete frequency	Locate the $N/2$ th item using cumulative frequency; for even N , average the $N/2$ th and $(N/2 + 1)$ th items if they are different.
Continuous frequency	Median = $L + \frac{(N/2 - c.f.)}{f} h$.

Exam Focus.**Median examples.**

Individual: For 2, 4, 6, 8, 10, 12, the middle values are 6 and 8, so median = $(6 + 8)/2 = 7$.

Discrete: For $x = 10, 20, 30, 40$ with $f = 2, 3, 4, 1$, the 5th item is 20 and the 6th is 30. Hence median = $(20 + 30)/2 = 25$.

Continuous: The median class is 10–20. With $L = 10$, $c.f. = 2$, $f = 3$, $h = 10$: median = $10 + [(5 - 2)/3]10 = 20$.

2.3 Mode

Mode is the value with the highest frequency.

Data	Formula / method
Individual	Value occurring most often; there may be no mode or more than one mode.
Discrete frequency	x corresponding to the largest f .
Continuous frequency	First identify the modal class (largest f), then $\text{Mode} = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} h$, where f_1 is modal-class frequency, f_0 previous, and f_2 next.

Exam Focus.**Mode examples.**

Individual: In 2, 4, 6, 8, 10, 12, every value occurs once, so there is no mode.

Discrete: For $x = 10, 20, 30, 40$ with $f = 2, 3, 4, 1$, the largest frequency is 4; mode = 30.

Continuous: The modal class is 20–30. With $L = 20$, $f_1 = 4$, $f_0 = 3$, $f_2 = 1$, $h = 10$: mode = $20 + [(4 - 3)/(4 - 3 - 1)]10 = 22.5$.

Do Not Confuse.

Mean uses every observation; median depends on position; mode depends on frequency. Do not use the grouped-data mode formula for individual data.

3. Measures of dispersion

Dispersion describes the spread of observations around a central value. A smaller dispersion means the observations are more closely clustered.

3.1 Range and coefficient of range

$$\text{Range} = L - S, \quad \text{Coefficient of range} = \frac{L - S}{L + S},$$

where L is the largest value and S is the smallest value.

Exam Focus.**Range examples.**

Individual: For 2, 4, 6, 8, 10, 12, range = $12 - 2 = 10$; coefficient = $(12 - 2)/(12 + 2) = 5/7$.

Discrete: For values 10 to 40, range = $40 - 10 = 30$; coefficient = $30/50 = 0.6$.

Continuous: For classes 0–10 to 30–40, range = $40 - 0 = 40$; coefficient = $40/40 = 1$. Use true boundaries if supplied.

3.2 Quartiles, IQR, and quartile deviation

$$\text{IQR} = Q_3 - Q_1, \quad QD = \frac{Q_3 - Q_1}{2}, \quad \text{Coefficient of } QD = \frac{Q_3 - Q_1}{Q_3 + Q_1}.$$

For grouped continuous data:

$$Q_1 = L + \frac{(N/4 - c.f.)}{f} h, \quad Q_3 = L + \frac{(3N/4 - c.f.)}{f} h.$$

Exam Focus.**Quartile, IQR, and QD examples.**

Individual: For 2, 4, 6, 8, 10, 12, $Q_1 = 4$, $Q_3 = 10$; $IQR = 6$, $QD = 3$, coefficient of QD = $6/14 = 3/7$.

Discrete: For $x = 10, 20, 30, 40$ with $f = 2, 3, 4, 1$, Q_1 is the 3rd item = 20 and Q_3 is the 8th item = 30; $IQR = 10$, $QD = 5$, coefficient = 0.2.

Continuous: $Q_1 = 10 + [(2.5 - 2)/3]10 = 11.667$ and $Q_3 = 20 + [(7.5 - 5)/4]10 = 26.25$. Thus $IQR = 14.583$ and $QD = 7.292$ approximately.

3.3 Mean deviation

Mean deviation is the average of absolute deviations from a chosen central value a (mean, median, or mode).

Data	Formula
Individual	$MD_a = \frac{\sum x - a }{n}$
Discrete frequency	$MD_a = \frac{\sum f x - a }{N}$
Continuous frequency	$MD_a = \frac{\sum f x - a }{N}$ using class marks.

Coefficient of mean deviation about a :

$$\text{Coefficient of MD} = \frac{MD_a}{a}.$$

Exam Focus.**Mean deviation examples about the mean.**

Individual: For 2, 4, 6, 8, 10, 12, $\bar{x} = 7$ and $\sum |x - 7| = 18$; $MD = 18/6 = 3$.

Discrete: For $x = 10, 20, 30, 40$ with $f = 2, 3, 4, 1$, $\bar{x} = 24$ and $\sum f|x - 24| = 80$; $MD = 80/10 = 8$.

Continuous: Using class marks 5, 15, 25, 35, $\bar{x} = 19$ and $\sum f|x - 19| = 80$; $MD = 80/10 = 8$. The grouped answer is approximate.

3.4 Standard deviation

Standard deviation is the positive square root of the mean squared deviation from the arithmetic mean.

Data	Direct formula
Individual	$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$
Discrete frequency	$\sigma = \sqrt{\frac{\sum f(x - \bar{x})^2}{N}}$
Continuous frequency	$\sigma = \sqrt{\frac{\sum f(x - \bar{x})^2}{N}}$, using class marks.

Shortcut formula

$$\sigma = \sqrt{\frac{\sum x^2}{n} - \bar{x}^2}, \quad \sigma = \sqrt{\frac{\sum fx^2}{N} - \bar{x}^2} \quad \text{for frequency data.}$$

Assumed-mean and step-deviation formulas

$$\sigma = \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2}, \quad d = x - A,$$

$$\sigma = h \sqrt{\frac{\sum fu^2}{N} - \left(\frac{\sum fu}{N}\right)^2}, \quad u = \frac{x - A}{h}.$$

Coefficient of variation:

$$CV = \frac{\sigma}{\bar{x}} \times 100\%.$$

Exam Focus.

Standard deviation examples.

Individual: For 2, 4, 6, 8, 10, 12, $\bar{x} = 7$ and $\sum(x - 7)^2 = 70$. Hence $\sigma = \sqrt{70/6} = 3.416$.

Discrete: For $x = 10, 20, 30, 40$ with $f = 2, 3, 4, 1$, $\bar{x} = 24$ and $\sum f(x - 24)^2 = 840$. Hence $\sigma = \sqrt{840/10} = 9.165$.

Continuous: For class marks 5, 15, 25, 35 with $f = 2, 3, 4, 1$, $\bar{x} = 19$ and $\sum f(x - 19)^2 = 840$. Hence $\sigma = \sqrt{840/10} = 9.165$ approximately.

Do Not Confuse.

Use N for population/descriptive standard deviation unless the question explicitly asks for sample standard deviation. For a sample estimate, use $s = \sqrt{\sum(x - \bar{x})^2 / (n - 1)}$ (or the frequency equivalent with $N - 1$).

4. Worked example A: individual data

Data: 2, 4, 6, 8, 10, 12.

Measure	Calculation
Mean	$\bar{x} = 42/6 = 7$.
Median	$n = 6$; median = $(6 + 8)/2 = 7$.
Mode	No mode: every value occurs once.
Range	$12 - 2 = 10$.
Coefficient of range	$(12 - 2)/(12 + 2) = 10/14 = 5/7$.
Quartiles	Lower half {2, 4, 6} gives $Q_1 = 4$; upper half {8, 10, 12} gives $Q_3 = 10$.
IQR and QD	$IQR = 10 - 4 = 6$; $QD = 6/2 = 3$.
Coefficient of QD	$(10 - 4)/(10 + 4) = 6/14 = 3/7$.
MD about mean	$\sum x - 7 = 5 + 3 + 1 + 1 + 3 + 5 = 18$; $MD = 18/6 = 3$.
Variance	$\sum (x - 7)^2 = 25 + 9 + 1 + 1 + 9 + 25 = 70$; $\sigma^2 = 70/6 = 11.667$.
Standard deviation	$\sigma = \sqrt{70/6} = 3.416$ approximately.
CV	$(3.416/7) \times 100 = 48.80\%$ approximately.

Exam Focus.

For individual data, arrange the values first for median and quartiles. Range uses only the smallest and largest values, while standard deviation uses every value.

5. Worked example B: discrete frequency data

x	f	fx	cf	$ x - 24 $	$f x - 24 $	$f(x - 24)^2$
10	2	20	2	14	28	392
20	3	60	5	4	12	48
30	4	120	9	6	24	144
40	1	40	10	16	16	256
Total	$N = 10$	$\sum fx = 240$			$\sum = 80$	$\sum = 840$

Measure	Calculation
Mean	$\bar{x} = \sum fx/N = 240/10 = 24$.
Median	5th item = 20, 6th item = 30; median = $(20 + 30)/2 = 25$.
Mode	Largest frequency is 4, so mode = 30.
Range	$40 - 10 = 30$.
Coefficient of range	$30/50 = 0.6$.
Quartiles	Q_1 is the 3rd item = 20; Q_3 is the 8th item = 30.
IQR and QD	$IQR = 10$; $QD = 5$; coefficient of QD = $10/50 = 0.2$.
MD about mean	$MD = \sum f x - 24 /N = 80/10 = 8$.
Variance	$\sigma^2 = \sum f(x - 24)^2/N = 840/10 = 84$.
Standard deviation	$\sigma = \sqrt{84} = 9.165$ approximately.
CV	$(9.165/24) \times 100 = 38.19\%$ approximately.

6. Worked example C: continuous frequency data

Classes and frequencies: 0–10: 2, 10–20: 3, 20–30: 4, 30–40: 1. Class marks are 5, 15, 25, 35 and $N = 10$.

Class	f	Class mark x	fx	cf	$f(x - 19)^2$
0–10	2	5	10	2	392
10–20	3	15	45	5	48
20–30	4	25	100	9	144
30–40	1	35	35	10	256
Total	10		190		840

Measure	Calculation
Mean	$\bar{x} = 190/10 = 19.$
Median	Median class 10–20; $L = 10$, $c.f. = 2$, $f = 3$, $h = 10$: $10 + (5 - 2)10/3 = 20.$
Mode	Modal class 20–30: $20 + (4 - 3)/(8 - 3 - 1) \times 10 = 22.5.$
Range	$40 - 0 = 40.$ If class boundaries are used, use the true lower and upper boundaries.
Coefficient of range	$(40 - 0)/(40 + 0) = 1.$
Q_1	Class 10–20: $10 + (2.5 - 2)10/3 = 11.667.$
Q_3	Class 20–30: $20 + (7.5 - 5)10/4 = 26.25.$
IQR and QD	$IQR = 26.25 - 11.667 = 14.583$; $QD = 7.292.$
MD about mean	Deviations from 19 are 14, 4, 6, 16; $\sum f x - 19 = 28 + 12 + 24 + 16 = 80$; $MD = 8.$
Standard deviation	$\sigma = \sqrt{840/10} = \sqrt{84} = 9.165$ approximately.
CV	$(9.165/19) \times 100 = 48.24\%$ approximately.

Remember.

For continuous distributions, median, quartiles, and mode use interpolation inside a class. Mean, MD, and SD use class marks, so they are approximate for grouped data.

7. Comparison and exam checks

Measure	Uses	Strength	Limitation
Mean	Every value.	Algebraically useful.	Affected by extreme values.
Median	Position.	Resistant to extreme values.	Less useful for algebra.
Mode	Highest frequency.	Works for qualitative data.	May be absent or non-unique.
Range	Extreme values only.	Fastest spread measure.	Very unstable.
IQR / QD	Middle 50%.	Resistant to extremes.	Ignores outer observations.
MD	Absolute deviations.	Easy to interpret.	Absolute values complicate algebra.
SD	Squared deviations.	Most widely used; algebraically strong.	Sensitive to extremes.

Do Not Confuse.

Common errors: forgetting to multiply by f ; using class limits instead of class boundaries; choosing the wrong median/quartile class; using N in a sample-SD question; reporting variance instead of standard deviation; and rounding before the final step.

8. Last-minute formula page

MEAN

$$\bar{x} = \sum x/n; \quad \bar{\bar{x}} = \sum fx/N.$$

MEDIAN

Individual: middle position.

Grouped: $L + [(N/2 - c.f.)/f]h$.

MODE

Grouped: $L + [(f_1 - f_0)/(2f_1 - f_0 - f_2)]h$.

RANGE

$L - S$; coefficient $(L - S)/(L + S)$.

QUARTILES

Grouped: $Q_1 = L + [(N/4 - c.f.)/f]h$; $Q_3 = L + [(3N/4 - c.f.)/f]h$.

IQR AND QD

$$IQR = Q_3 - Q_1; \quad QD = (Q_3 - Q_1)/2.$$

MEAN DEVIATION

$$MD = \sum |x - a|/n; \text{ frequency: } MD = \sum f|x - a|/N.$$

STANDARD DEVIATION

$$\sigma = \sqrt{\sum (x - \bar{x})^2/n}; \quad \text{frequency: } \sigma = \sqrt{\sum f(x - \bar{x})^2/N}.$$

SHORTCUT SD

$$\sigma = \sqrt{\sum fx^2/N - \bar{x}^2}.$$

CV

$$CV = (\sigma/\bar{x}) \times 100\%.$$

CHECKS

$SD \geq 0$; $IQR \geq 0$; MD uses absolute values; grouped results are approximate.